GRADE 7 ACCELERATED · UNIT 5 · LESSON 4

Real-World Scale Drawings

MA.7.GR.1.5

I can use a scale such as ¼ in : 1 ft to find real lengths and areas from a drawing, and drawing lengths from real ones.

Warm-Up

1. A figure with area 9 cm^2 is scaled by k = 2. Find the copy's area.
2. A 5 cm side becomes 3 cm in the copy. Find the scale factor.
3. Find the product: 1.5×2.4
4. A car goes 90 miles in 2 hours. How far does it go per hour?

Warm-Up

1. A figure with area 9 cm^2 is scaled by k = 2. Find the copy's area.

   36 cm^2

2. A 5 cm side becomes 3 cm in the copy. Find the scale factor.

   k = 0.6

3. Find the product: 1.5×2.4 **3.6**

4. A car goes 90 miles in 2 hours. How far does it go per hour?

   45 miles per hour

Vocabulary

floor plan

a scale drawing of a room or building seen from above

scale (two units)

a ratio such as ¼ in : 1 ft — a drawing length and the real length it stands for

unit rate of a scale

the scale written per 1 drawing unit: ¼ in : 1 ft is 1 in : 4 ft

Notes

I. A scale is a rate: write it per 1 unit

$$\frac{1}{4}\text{ in} : 1\text{ ft} \quad \rightarrow \quad 1\text{ in} : 4\text{ ft}$$

Drawing × rate = real. Real ÷ rate = drawing.

Notes

II. Area: go to the real dimensions first

1 in : 4 ft



$$8 \text{ ft} \times 10 \text{ ft} = 80 \text{ ft}^2$$

Notes

III. Real to drawing is a division

$$14 \div 4 = 3.5 \text{ in}$$

Example 1

A floor plan is drawn at ¼ in : 1 ft. A room is 2.5 in by 3 in on the plan.

Find the room's real dimensions and area.

Example 1

$\frac{1}{4}$ in : 1 ft = 1 in : 4 ft

$2.5 \times 4 = 10$ $\quad$ $3 \times 4 = 12$

$10 \times 12 = 120$

Answer: 10 ft by 12 ft; 120 ft²

Your Turn

A floor plan is drawn at ¼ in : 1 ft. A room is 1.5 in by 4 in on the plan.

Find the room's real dimensions and area.

Your Turn

A floor plan is drawn at ¼ in : 1 ft. A room is 1.5 in by 4 in on the plan.

Find the room's real dimensions and area.

$1.5 \times 4 = 6$ $\quad$ $4 \times 4 = 16$

$6 \times 16 = 96$

Answer: 6 ft by 16 ft; 96 ft²

Example 2

A map is drawn at 1 cm : 50 km. Two towns are 7.4 cm apart on it. A bus travels at 100 km per hour.

How far apart are the towns, and how long does the bus take?

Example 2

$7.4 \times 50 = 370$

$370 \div 100 = 3.7$

Answer: 370 km; 3.7 hours

Your Turn

A map is drawn at 1 cm : 25 km. Two towns are 6.2 cm apart on it. A bus travels at 50 km per hour.

How far apart are the towns, and how long does the bus take?

Your Turn

A map is drawn at 1 cm : 25 km. Two towns are 6.2 cm apart on it. A bus travels at 50 km per hour.

How far apart are the towns, and how long does the bus take?

$$6.2 \times 25 = 155$$

$$155 \div 50 = 3.1$$

Answer: 155 km; 3.1 hours

Whiteboards · Question 1 of 9

A plan is drawn at ½ in : 1 ft. A wall is 4 in long on it.

How long is the real wall?

Answer it.

Whiteboards · Question 1 of 9

A plan is drawn at ½ in : 1 ft. A wall is 4 in long on it.

How long is the real wall?

$$\frac{1}{2} \text{ in} : 1 \text{ ft} = 1 \text{ in} : 2 \text{ ft}$$

$$4 \times 2 = 8$$

Answer: 8 ft

Whiteboards · Question 2 of 9

A plan is drawn at 1 in : 3 ft. A room is 2.5 in long on it.

How long is the real room?

Answer it.

Whiteboards · Question 2 of 9

A plan is drawn at 1 in : 3 ft. A room is 2.5 in long on it.

How long is the real room?

$$2.5 \times 3 = 7.5$$

Answer: 7.5 ft

Whiteboards · Question 3 of 9

A plan is drawn at 1 in : 3 ft. A real wall is 21 ft long.

How long is it on the plan?

Answer it.

Whiteboards · Question 3 of 9

A plan is drawn at 1 in : 3 ft. A real wall is 21 ft long.

How long is it on the plan?

$$21 \div 3 = 7$$

Answer: 7 in

Whiteboards · Question 4 of 9

A model car is built at 1 : 50 and is 8 cm long.

How long is the real car?

A. 4 m

B. 0.16 m

C. 400 m

D. 58 cm

Answer it.

Whiteboards · Question 4 of 9

A model car is built at 1 : 50 and is 8 cm long.

How long is the real car?

$$8 \times 50 = 400 \text{ cm}$$

$$400 \div 100 = 4 \text{ m}$$

Answer: A — 4 m

Whiteboards · Question 5 of 9

A plan is drawn at ¼ in : 1 ft. A real wall is 15 ft long.

How long is it on the plan?

Answer it.

Whiteboards · Question 5 of 9

A plan is drawn at ¼ in : 1 ft. A real wall is 15 ft long.

How long is it on the plan?

$1 \text{ in} : 4 \text{ ft} \qquad 15 \div 4 = 3.75$

Answer: 3.75 in

Whiteboards · Question 6 of 9

A plan is drawn at 1 in : 4 ft. A room is 2 in by 3 in on it.

What is the room's real area?

A. $24\ \text{ft}^2$

B. $96\ \text{ft}^2$

C. $40\ \text{ft}^2$

D. $20\ \text{ft}^2$

Answer it.

Whiteboards · Question 6 of 9

A plan is drawn at 1 in : 4 ft. A room is 2 in by 3 in on it.

What is the room's real area?

$$2 \times 4 = 8 \qquad 3 \times 4 = 12$$

$$8 \times 12 = 96$$

Answer: B — 96 ft²

Whiteboards · Question 7 of 9

A plane is 48 ft long. One model of it is built at 1 : 48, another at 1 : 72.

Which model is longer, and by how much?

Answer it.

Whiteboards · Question 7 of 9

A plane is 48 ft long. One model of it is built at 1 : 48, another at 1 : 72.

Which model is longer, and by how much?

$48 \times 12 = 576$ in

$576 \div 48 = 12$ $\quad$ $576 \div 72 = 8$

Answer: The 1 : 48 model, by 4 in (12 in against 8 in)

Whiteboards · Question 8 of 9

Mia, who is 12, enlarges her 9 in by 12 in poster into a mural at 1 in : 2 ft.

What are the mural's dimensions?

Answer it.

Whiteboards · Question 8 of 9

Mia, who is 12, enlarges her 9 in by 12 in poster into a mural at 1 in : 2 ft.

What are the mural's dimensions?

$$9 \times 2 = 18 \qquad 12 \times 2 = 24$$

Answer: 18 ft by 24 ft

Whiteboards · Question 9 of 9

Andrew's drawing shows a wall 3.5 in wide; the real wall is 26.25 ft wide. The drawn height is 4 in.

Find the real height and the wall's real area. Which step uses the scale factor?

Write your answer in sentences.

Whiteboards · Question 9 of 9

Andrew's drawing shows a wall 3.5 in wide; the real wall is 26.25 ft wide. The drawn height is 4 in.

Find the real height and the wall's real area. Which step uses the scale factor?

$26.25 \div 3.5 = 7.5$ ft per in $\qquad 4 \times 7.5 = 30$

$26.25 \times 30 = 787.5$

Answer: 30 ft high; 787.5 ft². The scale factor (7.5 ft per in) is the step from 4 in to 30 ft.

Independent Set

1. A plan is drawn at ½ in : 1 ft. A wall is 6 in long on it. Find the real length.

2. A plan is drawn at 1 in : 5 ft. A room is 3.2 in long on it. Find the real length.

3. A plan is drawn at 1 in : 5 ft. A real wall is 22 ft long. Find its length on the plan.

4. A plan is drawn at ¼ in : 1 ft. A room is 2 in by 3.5 in on it. Find the real dimensions and area.

5. A model boat is built at 1 : 40 and is 15 cm long. Find the real boat's length in metres.

6. A map is drawn at 1 cm : 20 km. Two towns are 8.5 cm apart on it. How far apart are they, and how long does a train at 85 km per hour take?

IXL

1. Open IXL and start today's skills.
2. Work on paper where the question needs work. The answer box does not show it.
3. Every skill listed is required, to a SmartScore of 67. Due at the start of the next class.

Today's skills — all required

Scale drawings: word problems — 84H

Scale drawings: scale factor word problems — KCM